

there is no backlight to add bulk to it. According to an announcement by LG, they plan to develop a 152cm (60-inch) ultra-high-definition (ultra-HD) flexible and transparent OLED panel with a transmittance of more than 40 per cent and a curvature radius of 100R, by 2017.

Of course, no race is complete without Apple in it! In 2015, Apple was awarded a patent (<http://tinyurl.com/zsjoea4>) by the USPTO for flexible, foldable electronic devices with force gesture control. According to *Apple Insider*, the patent describes a portable device, like an iPhone, that can be flexed, bent, folded or otherwise deformed without negatively impacting the sensitive internal components contained within. What makes this patent exciting is that it also describes flexible external parts, such as the chassis and cover glass, with pliable internal parts like batteries, circuit boards, displays and other electrical components.

Not only that, it also hints at an exciting new user interface using force sensors. Say, for example, users can squeeze the device to activate a command.

The patent also mentions bi-stable flex regions. This means the device can have two or more standard configurations. So what would Apple make with this technology? A phone that folds up like a hanky or maybe a phone that has displays on both sides? Well, we might have to wait for a few years to find out.

Self-repairing screens

Users with toddlers racing around the house would readily vote for this technology. Last year, a team of chemists led by Duncan Wass of Bristol University, England, revealed a technology that would enable smartphones and other screens to heal themselves.

Inspired by the way the body forms scabs to protect wounds, this new technology involves adding a healing agent, basically a chemical formula, to materials used for

developing display screens, or even other products like nail varnish or wall paints.

This healing agent is made of different carbon based chemicals and produces a sheet of millions of microscopic spheres. When there is a crack in the display screen, it breaks apart these hollow microspheres, and a liquid is released that moves into the newly-formed gap. A subsequent chemical reaction causes this liquid to harden and seal the crack automatically, without any intervention from the user.

At a top-level, it looks as if including this technology into existing products would only involve a small manufacturing tweak, of adding the

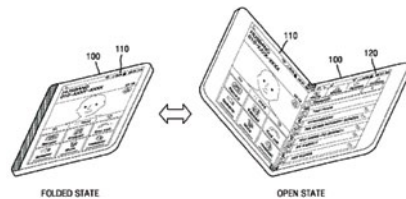


Fig. 1: Foldable device shown in Samsung's patent

healing agent to the material used. However, experts feel that new challenges might crop up when one starts delving into the details. The tech, for example, might require some adaptation to fit into cutting-edge flexible displays. However, it is definitely a good start and would save users a lot of money in replacing broken screens.

Displays that touch you

Displays seem to be incomplete without touch technology these days. As more and more devices fill the world, people want more natural forms of interacting with these. Since the past few years, there has been a lot of interest in haptic touchscreens, that is, screens that give you some form of sensory feedback.

Senseg is one of the pioneers in the field. Although not yet mainstream, their technology is lauded by experts and is constantly evolving. It lets the user feel reactions. Was the button-click accepted? Was there

an error? Is the call going through? All this can be felt through pulses or pressure exerted by the screen on the fingertips. Senseg is promoting the technology for use in automotive applications, where getting haptic feedback will reduce the necessity for the driver to look at the screen, thereby reducing distraction.

Bosch's haptic display, Neosense, apparently goes a step further, according to professional reviewers who tested it at CES 2016. According to reviews, Neosense lets you feel textures. Assume a system that has buttons designed with different textures, namely, coarse, silky, ribbed, etc. By moving your fingers over these, you can feel the different textures or patterns. This means that you can identify the required buttons simply by touching these.

Neosense also uses a vast array of pressure sensors to accurately differentiate between an intentional button press and brushing of fingers across the screen while searching for the right option.

Well, displays that let you feel textures would be a boon, especially for developers of learning systems and e-commerce portals. Imagine the day when customers can feel fabric on the screen while shopping for clothes!

Apple, too, has a bagful of haptic-feedback technologies like 3D Touch, Force Touch, Taptic Engine inside Apple Watch and vibration patterns for ringtones and notifications. However, haptic tech firm Immersion has filed lawsuits against Apple claiming that these infringe on Immersion's patents.

Will graphene and graphics get along

Since its discovery in the early 2000s, graphene has been touted as a magic material that has the potential to disrupt many an industry. It is basically a single-atom thick layer of carbon, achieved by shaving off layer after layer off graphite. Scientists propose innumerable uses for graphene, right from optics to semiconductors.